

did the military. Surgeons used HMDs to practice surgeries in super speciality hospitals, pilots ran simulations for flight training, and few architectural big wigs made virtual walkthroughs of their drawing board designs.

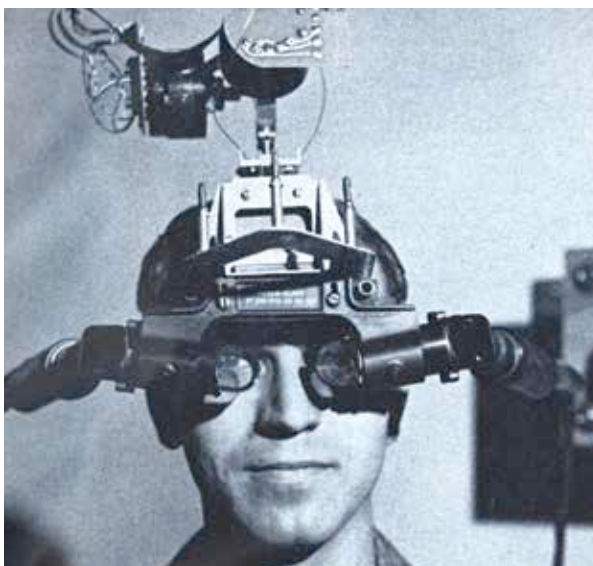
Soon the biggest names in gaming got the memo and experimented the best way they could. The mid-90s saw the release of Nintendo's Virtual Boy – a console that was almost unanimously declared a disaster. Users complained of nausea and the graphics just didn't look better than the gimmick that it turned out to be. Some would say they reached for the moon too soon. The supporting technology was just not available at that time.

Reality is a persistent b**ch. And it wasn't until 2009 that it met its match in the form of a plucky 16 year old kid named Palmer Luckey.

THE RIFT IS BRIDGED

Sitting in his garage, Palmer Luckey was fixing broken iPhones and investing whatever money he was earning to acquire high end gaming equipment and amassing a collection of old VR equipment through auctions. All the equipment he got his hands on only served to highlight how “not there” the technology was. It was plagued with lag i.e. a delay between the user moving his head and the actual visual panning in sync. This is one of the main things that causes nausea because your brain feels out of whack when deciphering visual stimulus. If it wasn't lag then the head mounted displays were too bulky to be of any practical use or the field of vision was too narrow to make them any good.

It was when he took apart these VR systems that he realised what the problem was. Many of these systems were using complicated lenses to warp the image to create the illusion of immersion. Palmer figured he could use off the shelf lenses and use software to distort the image to his need. Another fortuitous occurrence was that by this time smartphone displays had become quite advanced and quite cheap, thanks to their ubiquity and manufacturers trying to outdo each other with every given itera-



The Sword of Damocles hung over the user's head

tion. Palmer then used these smartphone parts to create the first prototype of what he called the Oculus Rift.

The experience that this little prototype delivered was far more lifelike and truly immersive. With motion tracking sensors mapping the user's head movement you could look around your virtually created environment. Soon this would expand to the possibility of inspecting objects by leaning in closer, even peering around walls. The chasm between the real and virtual was that much closer to being bridged.

Many iterations later Luckey was lucky enough to get his first big break when cult game developer John Carmack took an interest in his little hobby project. He took the Rift to the 2012 E3 conference and wowed audiences with

the promise of this window to the unreal – something journalists and audience members had never experienced. The rest of the story we all know – a Kickstarter campaign that aimed for a humble \$250,000 in funding but ended up netting \$2.4 million within days. This culminating in Facebook buying out Luckey's fledgling company lock and stock for a cool, \$2 billion.

THE UNREAL AWAITS

The Oculus Rift is no longer just a product. Sure it's a device, but it's also a catalyst spearheading an entire movement – a sort of steady march towards the unreal.

It has already inspired me-too devices like Sony's Project Morpheus and Samsung's Gear VR. Clearly the collective tech itself is on a steep curve towards improvement. The retail version of the Rift is slated to have a resolution higher than 1,920 by 1,080 pixels per eye (check the box on how it works) which would make the experience even more immersive than what we've experienced so far.

The mere fact that Facebook has entered the fray is proof enough that VR is now poised for widespread adoption not just in gaming but even other forms of social interaction. Imagine virtual experiences tailor made to your needs. Imagine travellers checking out hotel rooms by walking around in them before booking without ever leaving the comfort of their homes. And why limit yourself to worldly experiences? Reportedly

VR IN POP CULTURE: QUESTWORLD

Adventure was given a different albeit interesting meaning with The Real Adventures of Jonny Quest, an animation gem of the 90's. The most fascinating episodes were the ones that took place in the virtual world of QuestWorld which showed the protagonist and his followers inside a computer generated three-dimensional world meddling with stuff while on an adventure. Originally created for simulating experiments and as a research platform, QuestWorld was a very real virtual world. The reality of this virtual realm was such that fatal harm done to an avatar in the QuestWorld would actually be damaging to the user's body and brain (The Matrix?). QuestWorld is a perfect example of simulated reality and how it can be useful for experimentation without harming the real world (but you're not exempt).

